

Supplemental Material for Networks, Phillips Curves, and Monetary Policy

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1 Exogenous cost-push shocks

This section extends the model presented in the main text to allow for exogenous sector-level cost-push shocks, modeled as a change in the producers' desired markups. I denote these changes by an $N \times 1$ vector $\boldsymbol{\mu}^D$. Lemma (7) derives sectoral inflation rates and the Phillips curve.

Lemma 7. *Sectoral Phillips curves are given by*

$$\begin{aligned} \boldsymbol{\pi}_t = & \rho (I - \mathcal{V}) \mathbb{E} \boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi) \tilde{y}_t \\ & - \mathcal{V} \boldsymbol{\chi}_t + (I - \mathcal{V}) \Delta (I - \Delta)^{-1} \boldsymbol{\mu}_t^D \end{aligned} \quad (75)$$

The “divine coincidence” Phillips curve is

$$\pi_t^{DC} = \rho \mathbb{E} \pi_{t+1}^{DC} + (\gamma + \varphi) \tilde{y}_t + \boldsymbol{\lambda}^T \boldsymbol{\mu}_t^D \quad (76)$$

while the consumer-price Phillips curve is

$$\pi_t^C = \beta^T (I - \mathcal{V}) \rho \mathbb{E} \pi_{t+1} + \kappa^C \tilde{y}_t + u_t^C + v_t^C \quad (77)$$

where

$$v_t^C = \frac{\boldsymbol{\beta}^T \Delta (I - \Omega \Delta)^{-1}}{1 - \boldsymbol{\beta}^T \Delta (I - \Omega \Delta)^{-1} \boldsymbol{\alpha}} \boldsymbol{\mu}_t^D$$

and u_t^C is the same as in Corollary 2 .

Like in the baseline model, the central bank faces a worse trade-off after a cost-push shock than after a negative productivity shock of the same size (i.e. $\boldsymbol{\lambda}^T d \log \mathbf{A}_t = -\boldsymbol{\lambda}^T \boldsymbol{\mu}_t^D$). We can see this from the inequality

$$\frac{u_t^C}{\boldsymbol{\lambda}^T d \log \mathbf{A}_t} < \frac{v_t^C}{\boldsymbol{\lambda}^T \boldsymbol{\mu}_t^D}$$

This is because natural output does not change under the cost-push shock, but firms would still like to increase their price. As a result, stabilizing the divine coincidence inflation index requires lowering the output gap proportionately to the sales-weighted sum of sectoral cost-push shocks. Lemma (8) solves for the optimal policy response in the static economy ($\rho = 0$).

Lemma 8. *Under the optimal policy, the static inflation target in equation (31) takes value*

$$\pi^* = (\boldsymbol{\lambda}^T - \boldsymbol{\mathcal{B}}^T (I - \Delta) \Delta^{-1} \boldsymbol{\mathcal{L}}^{across}) \boldsymbol{\mu}^D \quad (78)$$

The first term of the policy target (78) captures the worse trade-off faced by monetary policy under the cost-push shock compared to a productivity shock, because natural output has not fallen. The central bank trades off the increase in inflation with an output loss, so that the optimal output gap is lower than after an equally-sized negative productivity shock. The second term in equation (78) reflects the fact that changes in sectoral markups also leads to misallocation, and managing the output gap can potentially correct this.

2 Complements to Section 6

2.1 Slope of the Phillips curve over time

2.1.1 Sectoral contributions

Figure 1 in Section 6.2.1 of the main text shows that the Phillips curve due to changes in the input-output structure versus changes in the composition of the consumption basket, period-by-period. In a similar way, the change from the beginning to the end of the sample can be decomposed as:

$$\delta_{\beta} \boldsymbol{\alpha}^{2017} - \delta_{\beta} \boldsymbol{\alpha}^{1947} = \frac{\boldsymbol{\beta}_{1947}^T + \boldsymbol{\beta}_{2017}^T}{2} (\mathbf{PT}_{2017} - \mathbf{PT}_{1947}) +$$

$$+ (\beta_{2017}^T - \beta_{1947}^T) \frac{\mathbf{PT}_{1947} + \mathbf{PT}_{2017}}{2} \quad (79)$$

where

$$\mathbf{PT}_t \equiv \Delta (I - \Omega_t \Delta)^{-1} \alpha_t$$

Consistent with Figure 1, the input shares component (second line of equation (79)) accounts for 65% of the overall decline. This decomposition captures the aggregate effect of changes in consumption and input shares. Figure 2.1.1 breaks it down into sector-level components.

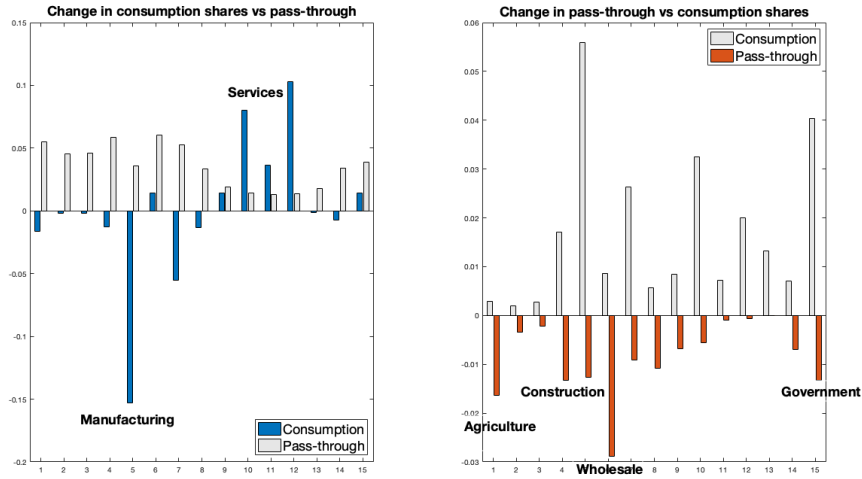


Figure 7: Left panel: change in consumption shares (in color) and average wage pass-through (shaded). Right panel: change in pass-through (in color) and average consumption shares (shaded).

The grey bars on the left panel represent the average pass-through $\frac{1}{2} (PT_{i,1947} + PT_{i,2017})$ for each sector i , while on the right panel they represent the average consumption share $\frac{1}{2} (\beta_{i,1947} + \beta_{i,2017})$. The bars in color represent changes in sectoral consumption shares $\beta_{i,2017}^T - \beta_{i,1947}^T$ and pass-through $PT_{i,2017} - PT_{i,1947}$. From the left panel we see that consumption shifted away from manufacturing (which has high pass-through) towards services (which have lower pass-through). The right panel shows that the pass-through decreased in all sectors, and more so in sectors with high consumption share (such as construction, manufacturing and government). Both channels lead to a flatter Phillips curve, although quantitatively the change in pass-through accounts for most of the effect.

2.1.2 Non-labor component of value added

Figure 1 in the main text is based on the benchmark calibration, which nets out the non-labor component of value added from total costs, implicitly attributing it to markups. In an alternative calibration, we treat the entirety of value added as payments to primary factors. The implications for the flattening of the Phillips curve are sensitive to this assumption. In fact, the non-labor share of value added has increased substantially over time, thereby offsetting the flattening of the Phillips curve coming from changes in the input-output structure. Figure 2.1.2 plots the slopes of the CPI and DC Phillips curves over time implied by this alternative calibration.

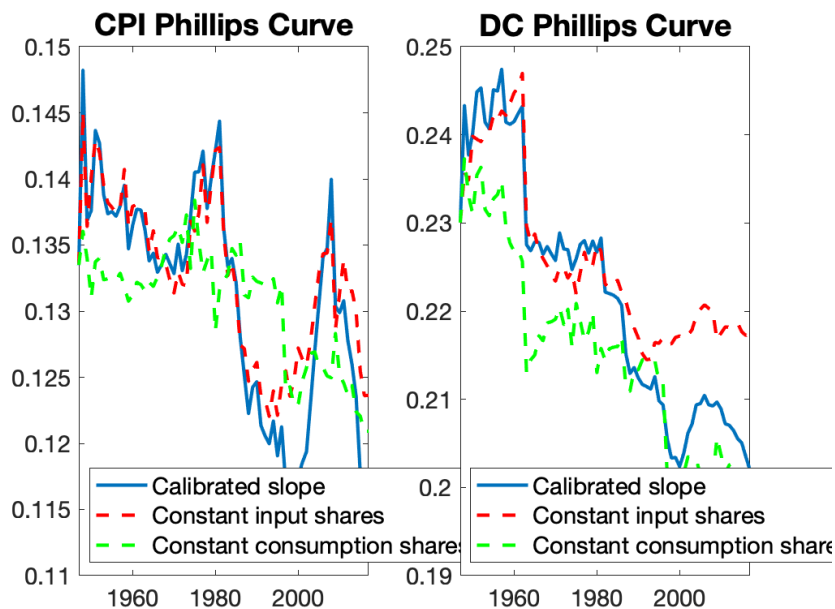


Figure 8: Phillips curve slopes over time, alternative calibration

2.2 Within- versus across-sector misallocation

Section 5.2 shows that the optimal policy is very similar to closing the output gap. Therefore under this policy the welfare loss from labor-leisure distortions is very small. The largest part of the welfare loss then comes from misallocation, either within or across sectors. Figure 9 compares the relative magnitude of these components. The three sets of bars in the figure correspond to different policy rules (optimal policy, output gap targeting and consumer price targeting). The within-sector component dominates, because we assumed a much higher elasticity of substitution within sectors

than across sectors.

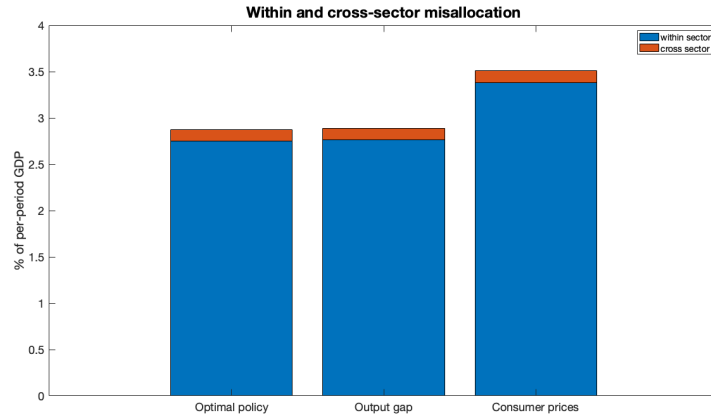


Figure 9: Substitution elasticities: $\epsilon = 8$, $\sigma = 0.9$, $\theta_L = 0.5$, $\theta = 0.001$

2.3 Monetary non-neutrality

The slope of the Phillips curve is tightly connected with the degree of monetary non-neutrality. We say that an economy has more monetary non-neutrality when output responds more, and inflation responds less, to exogenous changes in the nominal interest rate. Intuitively, a fall in the nominal rate stimulates demand, thereby raising output and inflation. The increase in inflation counteracts the initial demand expansion by reducing real incomes, and this mechanism is weaker when the Phillips curve is flatter. Therefore, we expect more monetary non-neutrality in economies that use more intermediate inputs.

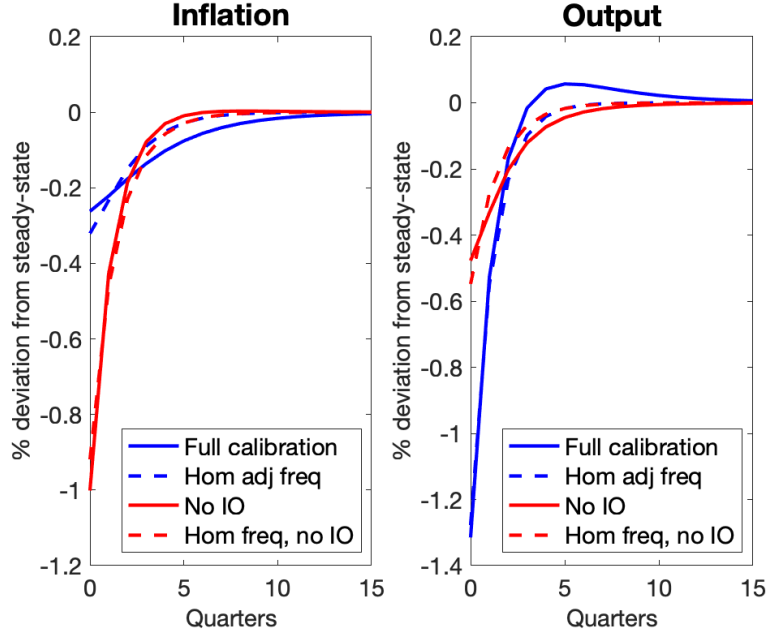


Figure 10: Impulse response to a 10 bp nominal rate shock, with persistence 0.5, and Taylor rule $i_{t+1} = r_{t+1}^{nat} + \zeta_{\pi}\pi_t^C + \zeta_y\tilde{y}_t + \epsilon_{t+1}$ with $\zeta_{\pi} = 1.24$ and $\zeta_y = .33/12$.

Figure 10 plots impulse-responses of inflation and output to a 1% change in the nominal rate, comparing economies with and without input-output linkages, and with uniform versus heterogeneous sectoral price adjustment probabilities. Like in Table III, accounting for intermediate inputs significantly increases monetary non-neutrality, while heterogeneity in price adjustment frequencies has no significant effect.

3 Additional empirical results

3.1 Construction of the divine coincidence index

The time series for the divine coincidence index π^{DC} is constructed by aggregating sector-level price series provided by the BLS PPI dataset. In this dataset the sample period varies across sectors. Most manufacturing series are available from the mid-1980s, while most service series are available from 2006 onwards. Out of the 405 sectors in the BEA classification, 172 have an incomplete price series in the BLS dataset, and 67 are missing.

Of the missing series, 40 can be obtained from sector-level data underlying the PCE, using the concordance table between NAICS sectors and PCE series provided

by the BEA. We used this same PCE data, which is available from 1960, to extend the incomplete price series further back in time. To do so, each incomplete PPI series is regressed on disaggregated (338 sectors) PCE components using the Lasso method, for the period in which both are available. This allows us to obtain a combination of PCE components that is the best predictor of each incomplete PPI series, which is then used as a proxy for the actual series in the periods when the PPI data is incomplete. The method yields very accurate predictions in out-of-sample tests.

Tables (11) and (12) report detail on the identity and weight of missing and incomplete series in the PPI dataset.

	Weight in DC	Added?
Oilseed farming	4.00	No
Funds, trusts, and other financial vehicles	2.02	Yes
Management of companies and enterprises	0.29	No
Sound recording industries	0.22	Yes
Elementary and secondary schools	0.20	Yes
Monetary authorities and depository credit intermediation	0.18	Yes
State and local government hospitals and health services	0.13	No
State and local government passenger transit	0.12	No
Other educational services	0.12	Yes
Motion picture and video industries	0.12	Yes
Transit and ground passenger transportation	0.10	Yes
Limited-service restaurants	0.10	Yes
Federal general government (nondefense)	0.09	No
Full-service restaurants	0.08	Yes
Promoters of performing arts and sports and agents for public figures	0.07	Yes

Table 11: Weights of top-15 missing series in DC (in %)

	Weight in DC	Initial date
Employment services	0.85	09/01/1994
Management consulting services	0.55	09/01/2006
Insurance agencies, brokerages, and related activities	0.47	03/01/2003
Architectural, engineering, and related services	0.45	03/01/1997
Automotive equipment rental and leasing	0.45	03/01/1992
Custom computer programming services	0.41	09/01/2006
Specialized design services	0.37	03/01/1997
Nursing and community care facilities	0.36	03/01/2004
Services to buildings and dwellings	0.36	03/01/1995
Environmental and other technical consulting services	0.36	09/01/2006
Wireless telecommunications carriers (except satellite)	0.31	09/01/1993
Office administrative services	0.27	09/01/1994
Satellite, telecommunications resellers, and all other telecommunications	0.23	09/01/1993
Other computer related services, including facilities management	0.22	09/01/2006
Internet publishing and broadcasting and Web search portals	0.21	03/01/2010

Table 12: Weights of top-15 incomplete series in DC (in %)

3.2 Full-sample regressions

Main Specification Tables 13, 14 and 15 report additional detail for the full-sample regressions in Section 7.2 of the main paper and Section 1.2 of the Online

		Unemp. Gap	Expect.	Constant	R^2
DC	Mich	-0.19 (0.03)	0.24 (0.07)	1.34 (0.23)	0.31
	SPF CPI	-0.17 (0.03)	0.06 (0.02)	1.89 (0.07)	0.31
	GMM	-0.16 (0.05)	-0.07 (0.22)	1.23 (0.43)	
CPI	Mich	-0.29 (0.07)	0.38 (0.21)	1.77 (0.63)	0.12
	SPF CPI	-0.26 (0.07)	0.1 (0.05)	2.62 (0.19)	0.12
	GMM	-0.12 (0.04)	0.51 (0.1)	0.92 (0.2)	
coreCPI	Mich	-0.25 (0.06)	1.02 (0.16)	-0.16 (0.5)	0.27
	SPF CPI	-0.16 (0.06)	0.20 (0.04)	2.36 (0.15)	0.19
	GMM	-0.04 (0.04)	0.46 (0.29)	0.30 (0.16)	
PCE	Mich	-0.2 (0.06)	0.34 (0.18)	1.41 (0.55)	0.08
	SPF CPI	-0.16 (0.06)	0.09 (0.05)	2.17 (0.16)	0.08
	GMM	-0.06 (0.02)	0.63 (0.06)	0.44 (0.07)	
corePCE	Mich	-0.16 (0.05)	0.7 (0.15)	0.29 (0.46)	0.17
	SPF CPI	-0.10 (0.05)	0.15 (0.04)	2.00 (0.14)	0.12
	GMM	-0.02 (0.01)	0.70 (0.12)	0.09 (0.03)	
PPI	Mich	-0.14 (0.24)	-2.76 (0.71)	10.64 (2.17)	0.11
	SPF CPI	-0.37 (0.25)	-0.4 (0.19)	3.42 (0.66)	0.04
	GMM	-0.20 (0.13)	0.54 (0.10)	0.98 (0.38)	

Table 13: Regressions of yearly inflation on CBO's unemployment gap, 1984-2018

Appendix.

		Output Gap	Expect.	Constant	R^2
DC	Mich	0.19 (0.02)	0.26 (0.07)	1.36 (0.22)	0.33
	SPF CPI	0.16 (0.02)	0.07 (0.02)	1.96 (0.07)	0.33
	GMM	-0.13 (0.03)	1.57 (0.12)	-0.91 (0.22)	
CPI	Mich	0.29 (0.07)	0.41 (0.21)	1.81 (0.62)	0.13
	SPF CPI	0.25 (0.06)	0.11 (0.05)	2.73 (0.19)	0.13
	GMM	0.07 (0.03)	0.66 (0.15)	0.63 (0.28)	
coreCPI	Mich	0.13 (0.05)	0.97 (0.17)	-0.03 (0.52)	0.2
	SPF CPI	0.04 (0.06)	0.21 (0.04)	2.26 (0.17)	0.15
	GMM	-0.03 (0.01)	1.04 (0.05)	-0.13 (0.04)	
PCE	Mich	0.2 (0.06)	0.36 (0.18)	1.43 (0.54)	0.09
	SPF CPI	0.17 (0.06)	0.09 (0.05)	2.25 (0.17)	0.09
	GMM	0.1 (0.02)	0.45 (0.09)	0.72 (0.11)	
corePCE	Mich	0.1 (0.05)	0.68 (0.15)	0.36 (0.47)	0.13
	SPF CPI	0.04 (0.05)	0.15 (0.04)	1.96 (0.15)	0.1
	GMM	0.03 (0.01)	0.56 (0.19)	0.16 (0.04)	
PPI	Mich	0.61 (0.22)	-2.42 (0.69)	10.26 (2.11)	0.16
	SPF CPI	0.82 (0.22)	-0.42 (0.18)	4.24 (0.68)	0.12
	GMM	0.9 (0.27)	-0.1 (0.31)	3.46 (1.03)	

Table 14: Regressions of yearly inflation on CBO's output gap, 1984-2018

		Unemp. Rate	Expect.	Constant	R^2
DC	Mich	-0.15 (0.03)	0.25 (0.08)	2.08 (0.27)	0.18
	SPF CPI	-0.13 (0.03)	0.08 (0.02)	2.5 (0.18)	0.21
	GMM	-0.11 (0.04)	0.09 (0.16)	1.50 (0.50)	
CPI	Mich	-0.17 (0.08)	0.34 (0.22)	2.66 (0.71)	0.04
	SPF CPI	-0.13 (0.07)	0.12 (0.05)	3.18 (0.48)	0.06
	GMM	-0.08 (0.03)	0.56 (0.08)	1.15 (0.28)	
coreCPI	Mich	-0.08 (0.06)	0.94 (0.18)	0.38 (0.57)	0.17
	SPF CPI	0.01 (0.06)	0.21 (0.04)	2.17 (0.39)	0.14
	GMM	-0.04 (0.04)	0.42 (0.29)	0.44 (0.31)	
PCE	Mich	-0.07 (0.07)	0.28 (0.19)	1.87 (0.61)	0.02
	SPF CPI	-0.05 (0.06)	0.1 (0.05)	2.3 (0.41)	0.04
	GMM	-0.04 (0.01)	0.65 (0.06)	0.58 (0.12)	
corePCE	Mich	-0.02 (0.06)	0.62 (0.16)	0.52 (0.51)	0.11
	SPF CPI	0.04 (0.05)	0.15 (0.04)	1.67 (0.35)	0.1
	GMM	-0.01 (0.01)	0.71 (0.11)	0.14 (0.06)	
PPI	Mich	-0.12 (0.25)	-2.75 (0.72)	11.21 (2.35)	0.11
	SPF CPI	-0.38 (0.25)	-0.37 (0.19)	5.33 (1.65)	0.04
	GMM	-0.24 (0.16)	0.53 (0.11)	2.29 (1.23)	

Table 15: Regressions of yearly inflation on the unemployment rate, 1984-2018

Endogeneity of monetary policy and omitted variable bias A well known issue with Phillips curve regressions is that monetary policy manages the output gap so as to counteract the effect of cost-push shocks (see Woodford (1994), Williams (2006), Mishkin (2007), Fitzgerald and Nicolini (2014), McLeay and Tenreyro (2019)). This creates a negative correlation between the independent variable (the output gap) and the error term (the cost-push shock).

This may affect our interpretation of the results in Section 7 (the input-output model predicts well the empirical slope of the Phillips curve, and the employment gap is more correlated with the divine coincidence index than with consumer inflation).

To address this issue, Figures 11, 12, 13 and 14 plot the estimated coefficients and Rsquareds when running Phillips curve regressions on simulated data, as a function of the volatility and persistence of monetary shocks and cost-push shocks. The simulated model produces the same results as we observe in the actual data when exogenous cost-push shocks to the producers' desired markups have low volatility and moderate persistence, and monetary shocks have low volatility and persistence.

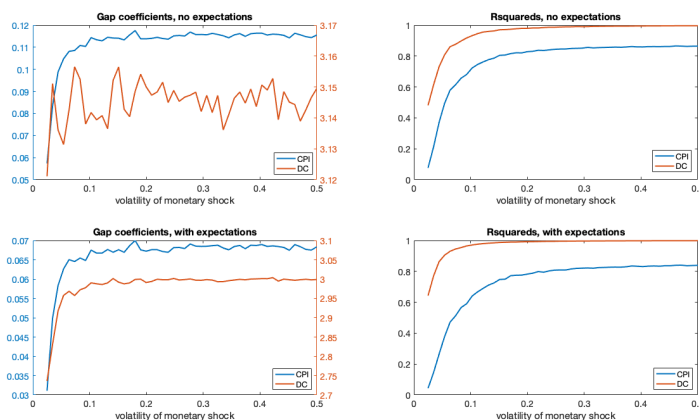


Figure 11:

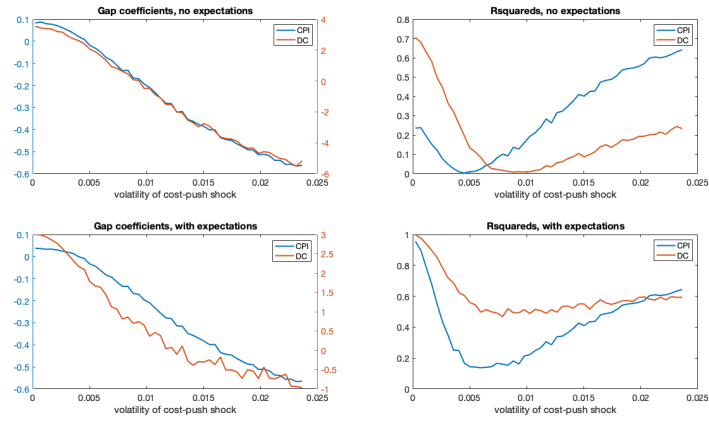


Figure 12:

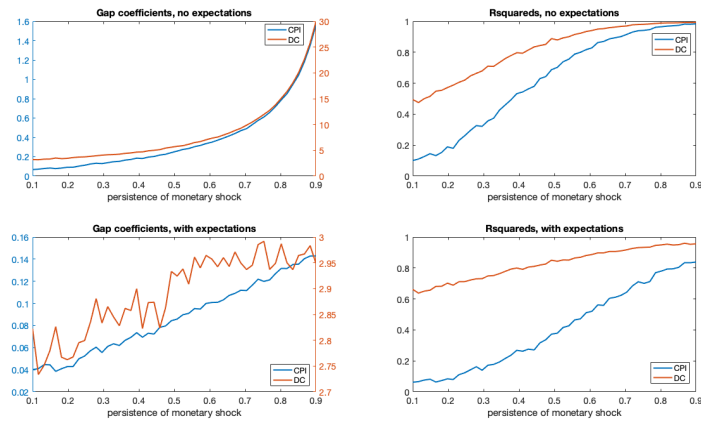


Figure 13:

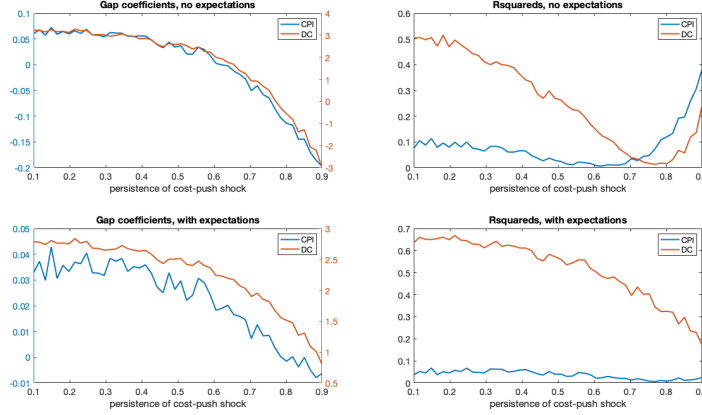


Figure 14:

3.3 Rolling regressions

The figures below provide additional detail for the rolling regressions introduced in Section 3.3. They plot estimated coefficients for each 20-year window (with confidence intervals), and average R-squareds over the sample. The years on the x -axis correspond to the middle of the estimation window.

We first report results for a specification with inflation expectations, as in equation (80), using the CBO unemployment gap, the CBO output gap and the unemployment rate as independent variables, and proxying for expectations with the SPF (first three figures) or the Michigan survey of Consumers (fourth to sixth figure).

$$\pi_t = \kappa \tilde{y}_t + \rho \mathbb{E} \pi_{t+1} + \epsilon_t \quad (80)$$

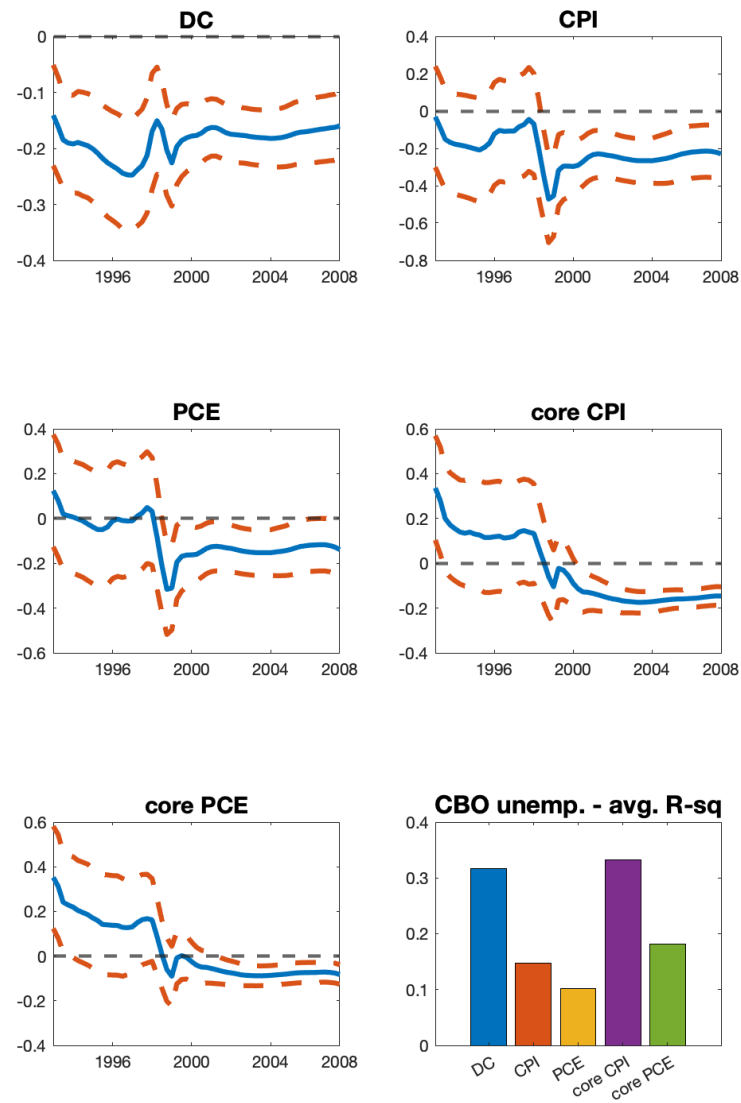


Figure 15: Rolling Phillips curve regressions, using the CBO unemployment gap and the SPF

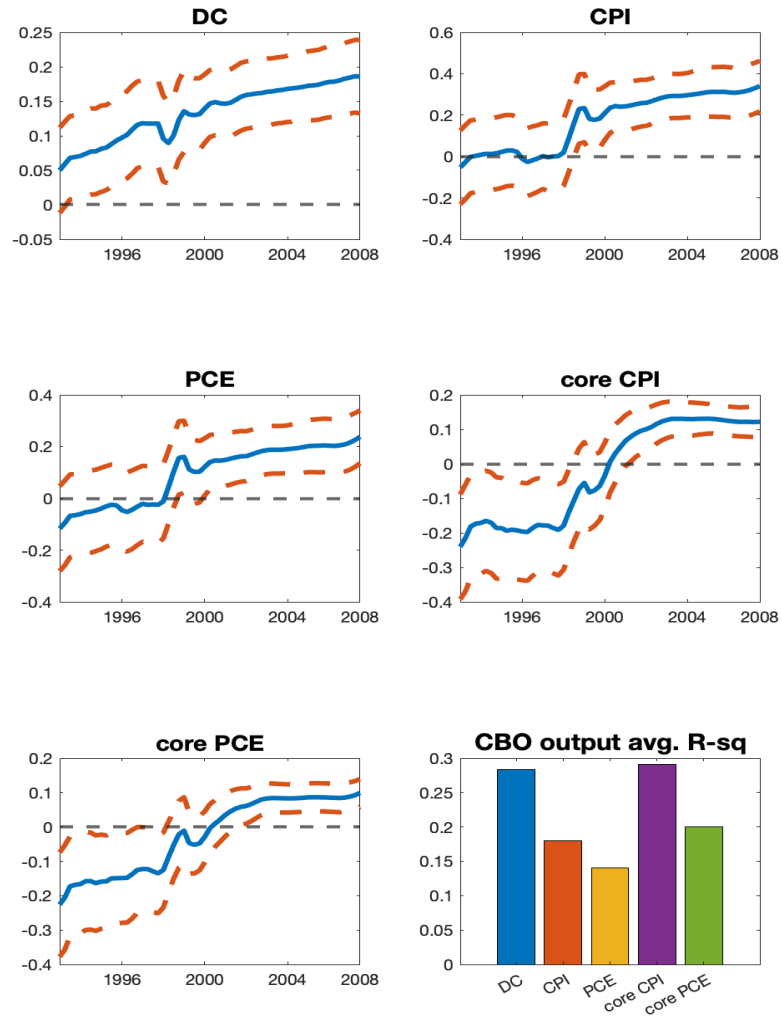


Figure 16: Rolling Phillips curve regressions, using the CBO output gap and the SPF

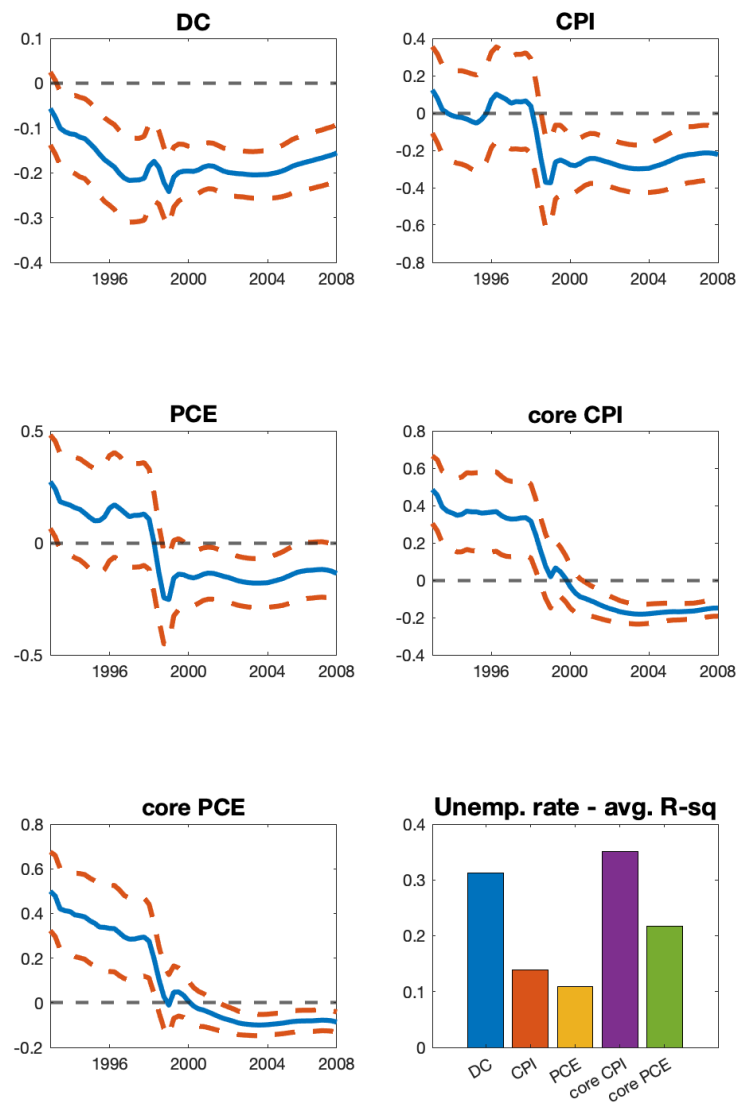


Figure 17: Rolling Phillips curve regressions, using the unemployment rate and the SPF

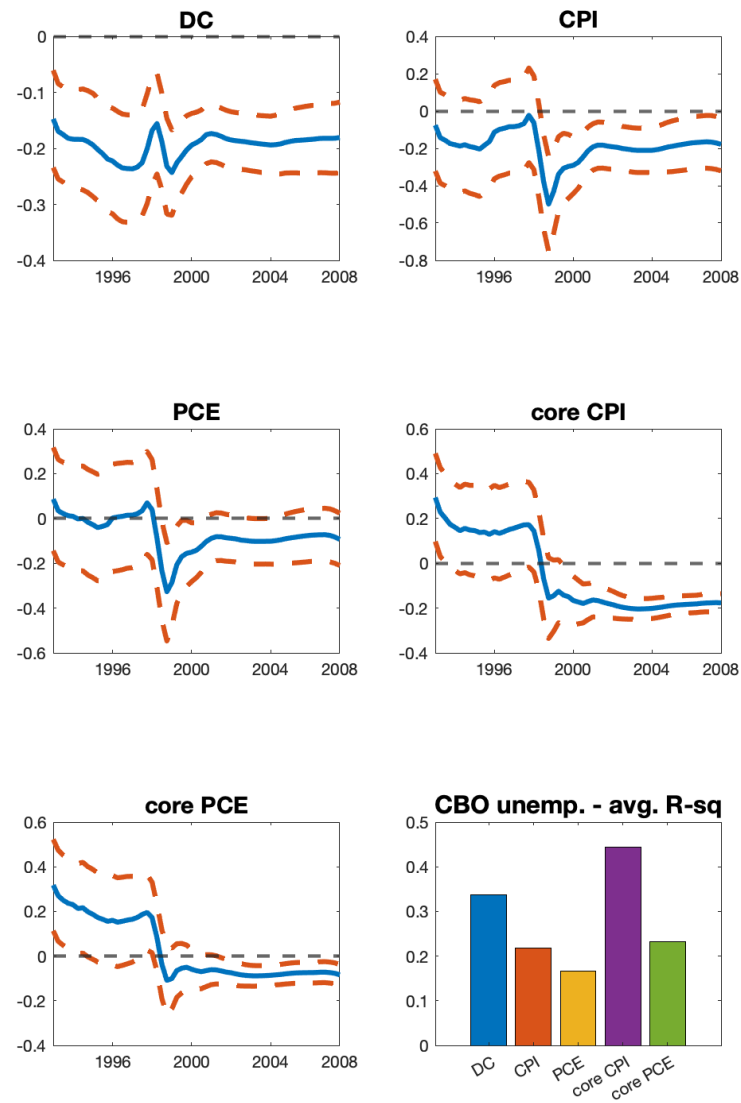


Figure 18: Rolling Phillips curve regressions, using the CBO unemployment gap and the Michigan Survey

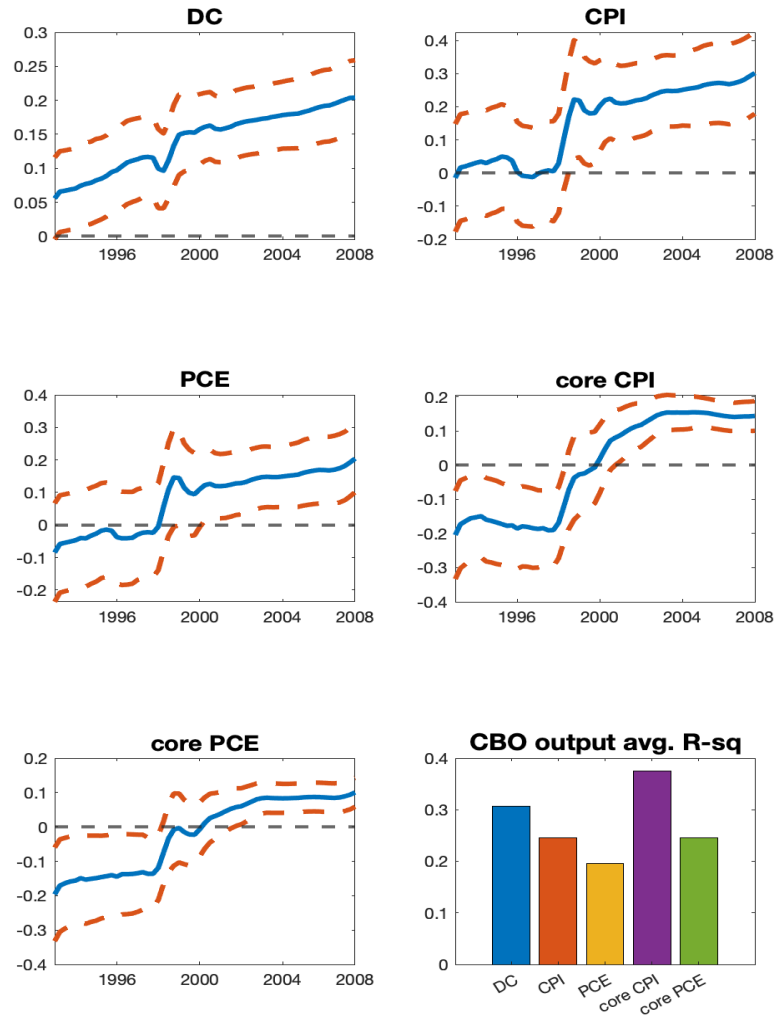


Figure 19: Rolling Phillips curve regressions, using the CBO output gap and the Michigan Survey

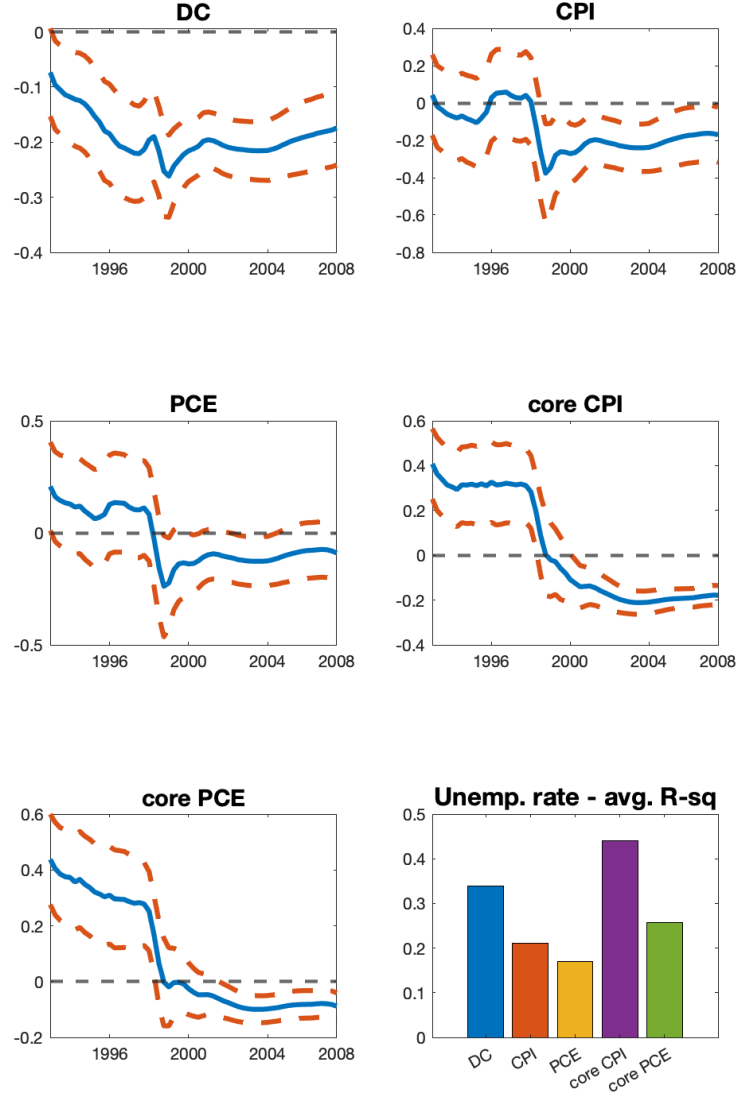


Figure 20: Rolling Phillips curve regressions, using the CBO unemployment gap and the SPF

The figures below report the same plots for a specification without inflation expectations

$$\pi_t = \kappa \tilde{y}_t + \epsilon_t$$

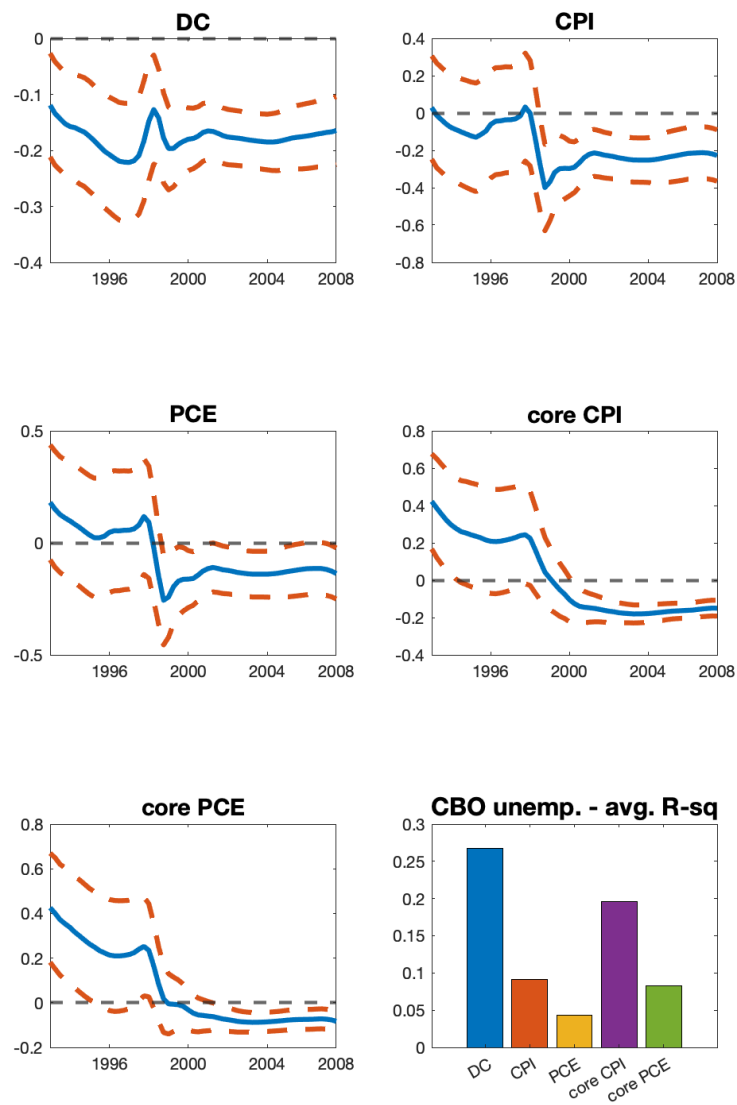


Figure 21: Rolling Phillips curve regressions, using the CBO unemployment gap

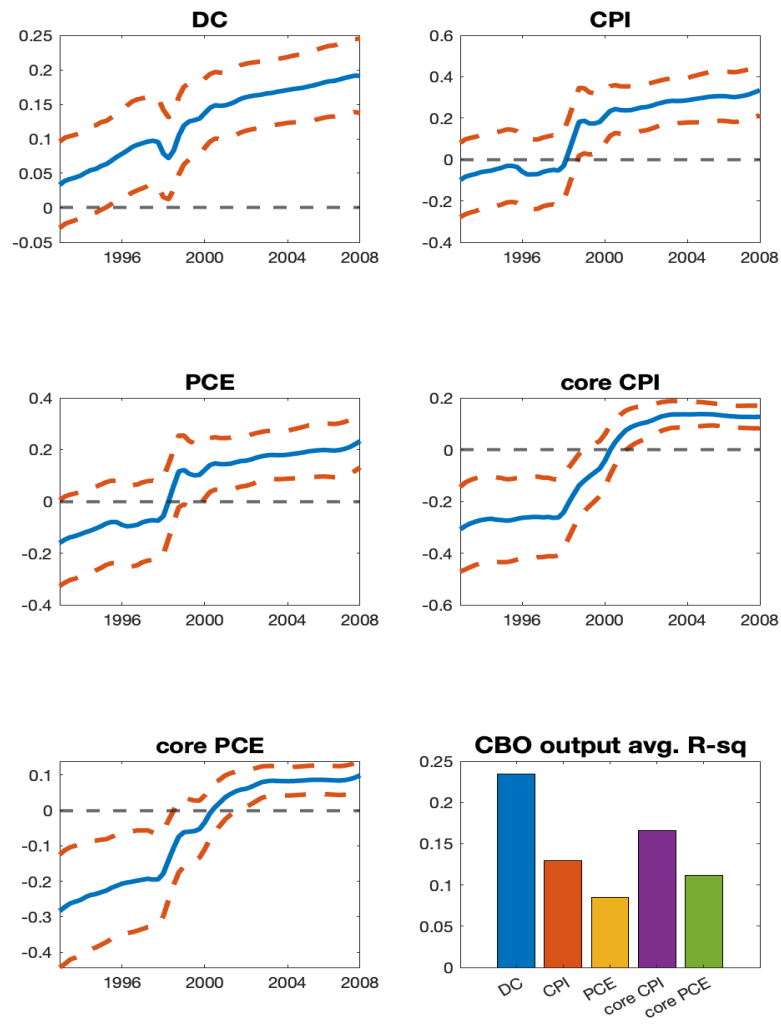


Figure 22: Rolling Phillips curve regressions, using the CBO output gap

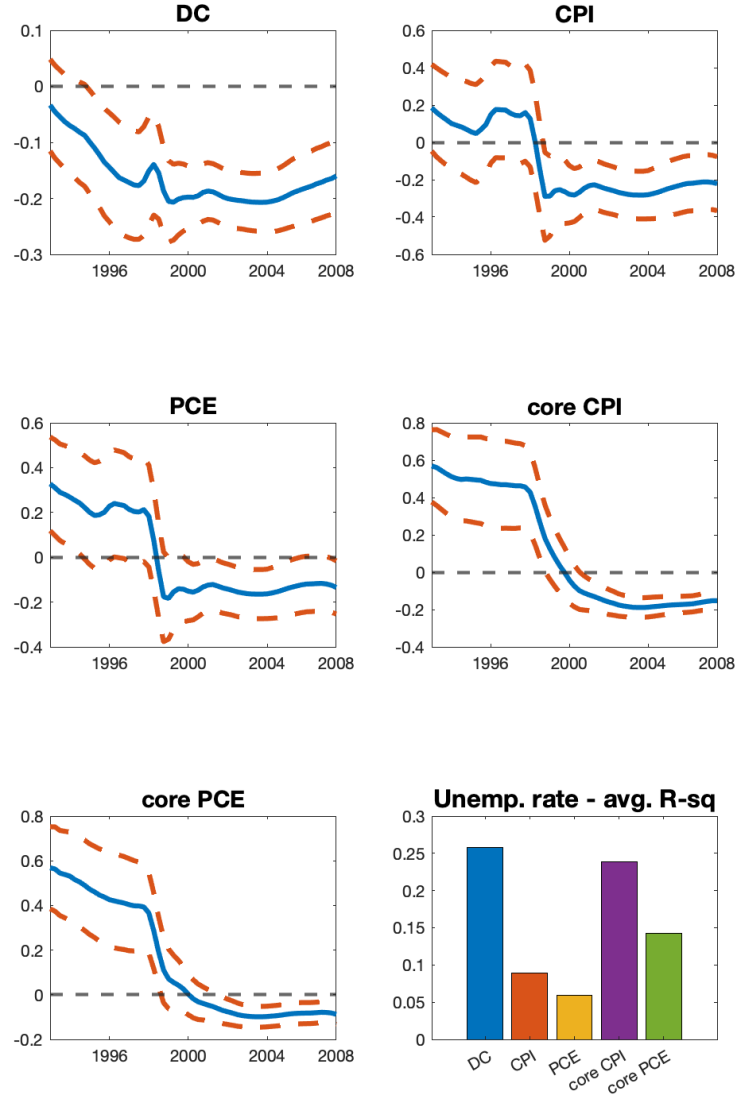


Figure 23: Rolling Phillips curve regressions, using the unemployment rate

3.4 Scatterplots

This section reports scatterplots of inflation and output gaps for all inflation and gap measures used in the regressions. Figures (3.4), (3.4) and (3.4) report scatterplots in levels, while Figures (3.4), (3.4) and (3.4) report scatterplots for inflation changes

versus gap levels.

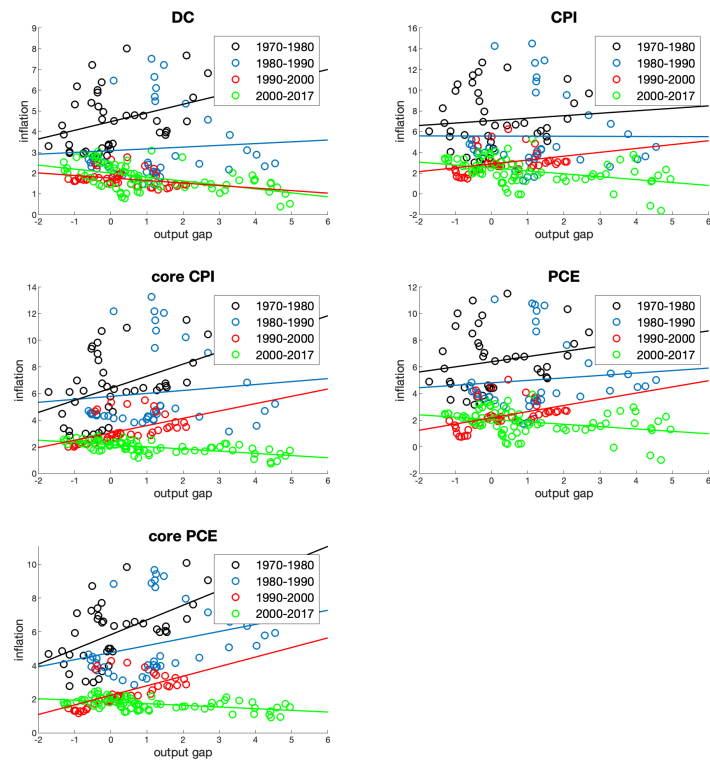


Figure 24: Inflation and unemployment gap

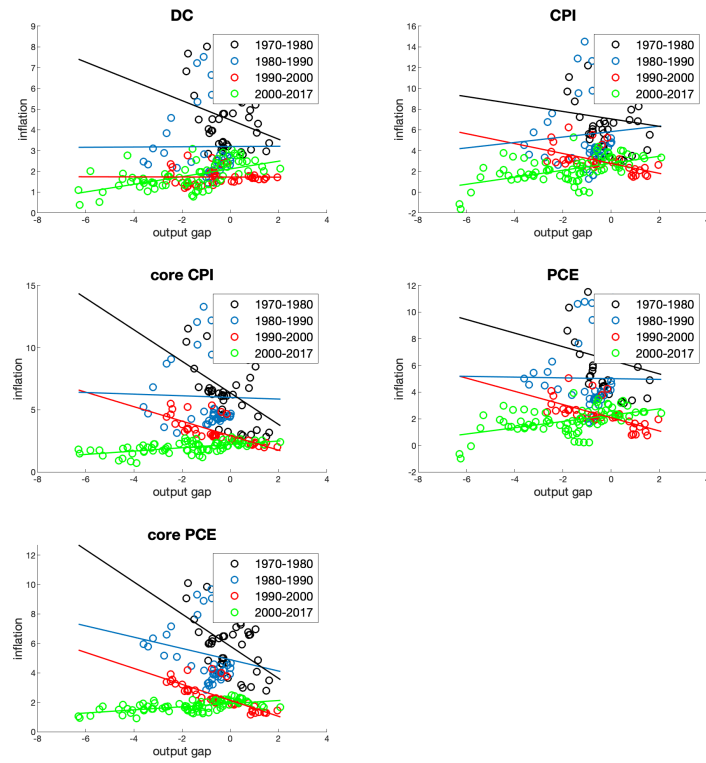


Figure 25: Inflation and output gap

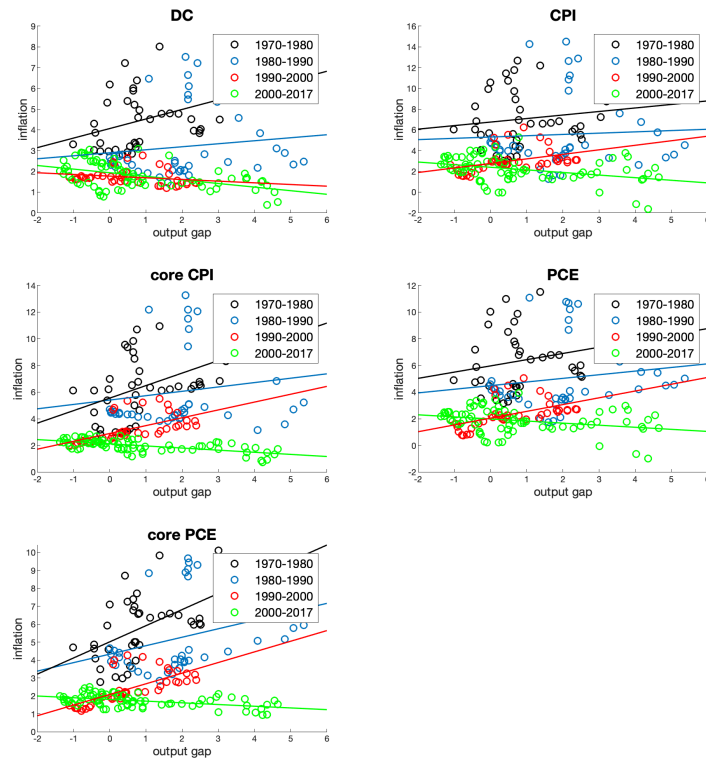


Figure 26: Inflation and unemployment rate

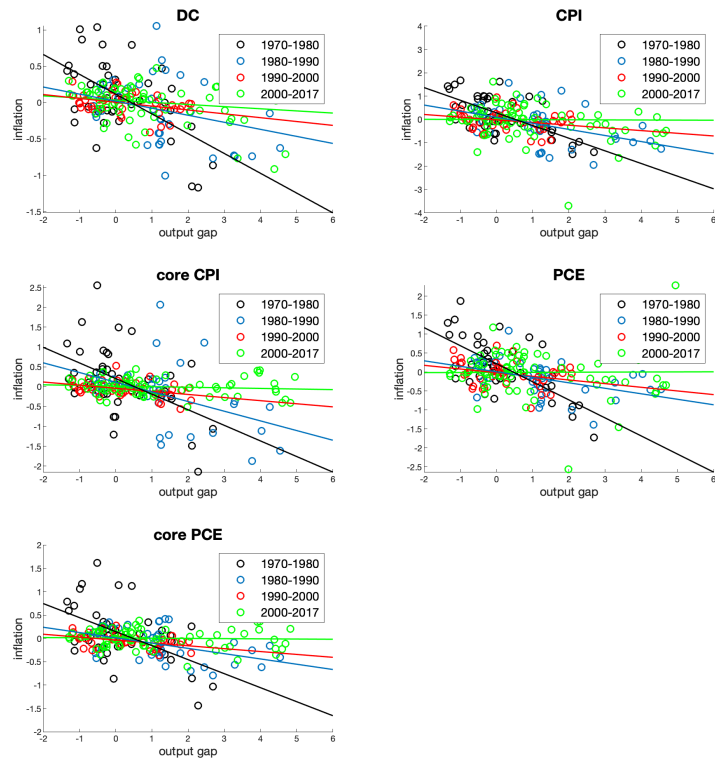


Figure 27: Inflation changes and unemployment gap

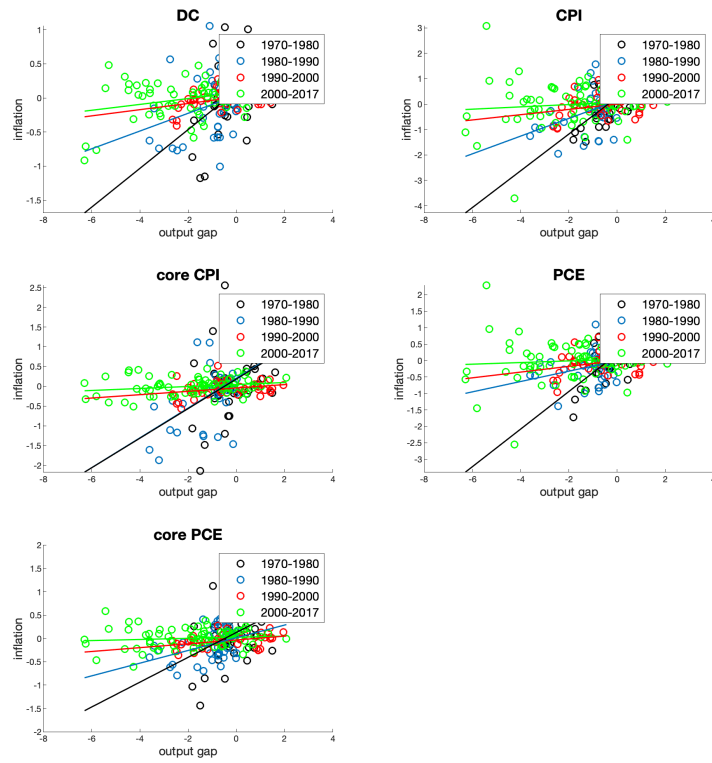


Figure 28: Inflation changes and output gap

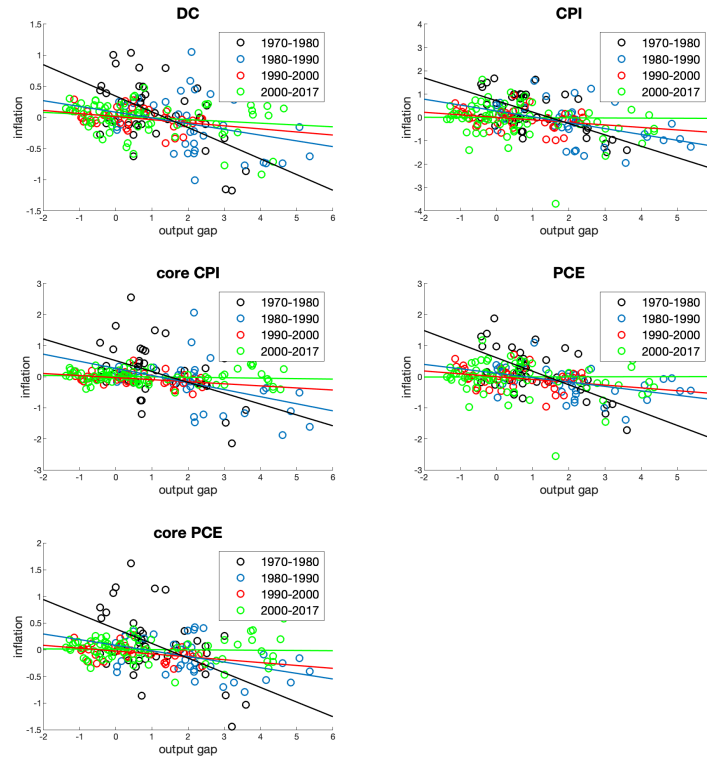


Figure 29: Inflation changes and unemployment rate

References

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